# Product Requirements Document

## Enterprise IT Support Agentic RAG Copilot

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| **Document Version** | 1.0.0 (Release Candidate) |
| **Customer / Org** | NovaRetail (Fictional 3,000-employee enterprise) |
| **Author / Lead Engineer** | Bittu Sharma (Forward Deployed Engineer - AI Systems) |
| **Target Audience** | Product Management, IT Leadership, Support Operations |
| **Approval Status** | Approved for Production Deployment |
| **Evaluation Score** | 88.9% Live Pass Rate \| 100% Mock Mode \| 100% Routing Accuracy |

## 1. Business Problem & Opportunity

NovaRetail employs 3,000 corporate and retail operations staff. The internal IT team maintains a comprehensive library of standard operating procedures: VPN configuration, identity and password policies, MFA requirements, software deployment rules, and laptop troubleshooting guides. Despite this documentation, employees submit over 1,200 repetitive support tickets monthly because internal knowledge is fragmented across file shares and wikis.

Traditional keyword search generates dozens of search hits without synthesis. Furthermore, standard off-the-shelf chatbots hallucinate policy details or fail entirely when questions depend on external vendor uptime status (e.g., Microsoft 365, AWS, Okta outages).

**BUSINESS OBJECTIVE:** The business objective is to deliver a trustworthy, self-correcting IT Support Copilot that reduces Tier-1 tickets by 40%+ by answering grounded questions from company documents first, evaluating evidence quality, and selectively querying external sources only when necessary.

## 2. User Personas & Core Journeys

| Persona | Role & Needs | Pain Points & Solution |
|---|---|---|
| **Enterprise Employee (Mark)** | Remote retail operations manager needing fast laptop/VPN resolution without jargon. | Frustrated by waiting 4+ hours for IT tickets. Needs step-by-step verified instructions immediately. |
| **IT Helpdesk Lead (Priya)** | Tier-2 Support engineer who handles escalated tickets and monitors system decisions. | Needs complete visibility into the AI's reasoning path, confidence %, and citations to verify answers. |
| **IT Systems Admin (David)** | Knowledge manager responsible for publishing updated corporate security policies. | Cannot wait for engineering release cycles to update bots. Needs zero-code drag-and-drop document indexing. |

# 3. Functional Requirements (FR)

• **FR-1: Query Routing Engine**: System must intelligently categorize incoming questions into KB queries (VPN, passwords, software) vs conversational greetings (direct response).

• **FR-2: Private Knowledge Retrieval**: System must perform dense vector similarity search across Pinecone enterprise namespace, retrieving Top-4 semantic chunks.

• **FR-3: Self-Reflective Evidence Grading**: A structured LLM grader must score retrieved internal evidence as GOOD or WEAK based on direct sufficiency.

• **FR-4: External Search Fallback**: When internal evidence is WEAK, workflow must invoke Tavily search to fetch live vendor guides without human intervention.

• **FR-5: Query Rewriter & Retry Guard**: If web evidence is insufficient, system rewrites technical query with technical keywords, guarded by max_retries=1 to prevent infinite loops.

• **FR-6: Dual-Tab User Experience**: Web application must provide an interactive Chat Stream tab and an automated Result & Insights tab showing full metrics.

• **FR-7: Document Ingestion Portal**: Authorized admins can upload PDF, DOCX, TXT, and Markdown files with token chunking and immediate vector indexing.

• **FR-8: Audit Logging & Security**: All questions, routed sources, and execution traces must be durably stored in SQLite for compliance auditability.

# 4. User Experience & Application KPIs

The user experience is designed with an enterprise dark-mode theme featuring a dual-tab dashboard:

• **[1] Chat Tab:** Fast conversation stream with numbered bubbles (#1 You, #1 Agent Response) and suggestion pills.

• **[2] Result & Insights Tab:** Auto-switched upon query completion, displaying Query ID, 96% Evidence Confidence, 95% Groundedness Match, Latency, Cited Sources, and Decision Trace steps.

• **KPI Targets:** Routing Accuracy >= 95% (Achieved 100%), Pass Rate >= 85% (Achieved 88.9% live, 100% mock), Average Latency < 15 seconds.